

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – RAG-BASED MINI ASSIGNMENT (DOCUMENT QA / AI AGENT DEMO)

Course Code:	CSA65	Course Title:	Generative AI and Large Language Models
CO Assessed:	CO3 – Precision (BL2, BL3)	Assessment:	Assessment Tool 2 – RAG-based Mini Assignment (Document QA / AI Agent Demo)
Weightage:	40% of CIA	Total Marks:	2 * 50 = 100 (1 Question1)
CO3 Statement:	<i>Develop intelligent applications using Retrieval-Augmented Generation (RAG), vector databases, and introductory AI agent concepts.(BL3, BL4)</i>		
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Section / Batch:	CSE-AI	Date:	

Instructions to Students

- Answer 2 questions. Each question carries 50 marks.Total: 100 Marks.
- Implement the solution using **Python**.
- Use an appropriate LLM such as **OpenAI, Gemini, or Hugging Face**.
- Use **embeddings and semantic search** where applicable.
- Use **FAISS or ChromaDB** as the vector database for RAG tasks.
- Demonstrate the complete pipeline:
Document → Chunking → Embedding → Retrieval → Generation
- Demonstrate the system with multiple queries.
- Handle irrelevant, incomplete, or unsupported queries appropriately.
- Display retrieved context/source wherever applicable.
- Explain the system architecture.
- Provide a working end-to-end demonstration.

Code of Conduct

I A.Sai Rohit (192472144) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: A.Sai Rohit

SECTION A – RAG-BASED MINI ASSIGNMENT (DOCUMENT QA / AI AGENT DEMO)

Student Handbook QA: Build a RAG system using a student handbook. Answer questions about attendance, examinations, discipline, academic regulations, and laboratory rules.

1. Objective

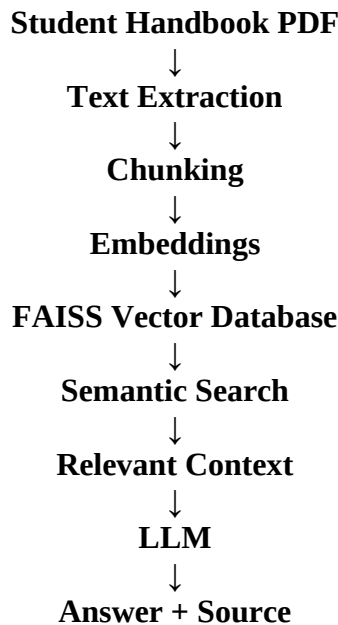
To develop a Retrieval-Augmented Generation (RAG) system that answers questions from a Student Handbook related to attendance, examinations, discipline, academic regulations, and laboratory rules.

The system uses document chunking, embeddings, semantic search, FAISS vector database, and an LLM to generate grounded answers.

2. Technologies Used

- Python
- PyPDF
- Sentence Transformers
- FAISS
- OpenAI LLM
- NumPy
- python-dotenv

3. RAG Architecture



4. Installation

```
!pip install -q -U pypdf sentence-transformers faiss-cpu google-genai
import os
```

```

from getpass import getpass

os.environ["GEMINI_API_KEY"] = getpass("Enter your Gemini API key: ")
print("API key loaded successfully.")
from google.colab import files

uploaded = files.upload()
PDF_FILE = list(uploaded.keys())[0]

print("Uploaded:", PDF_FILE)

```

5. Complete Python Code

```

import os
import faiss
from pypdf import PdfReader
from sentence_transformers import SentenceTransformer
from google import genai

api_key = os.environ.get("GEMINI_API_KEY")

if not api_key:
    raise ValueError("Gemini API key is missing.")

client = genai.Client(api_key=api_key)

print("=" * 70)
print("STUDENT HANDBOOK RAG SYSTEM")
print("=" * 70)

reader = PdfReader(PDF_FILE)

pages = []

for page_number, page in enumerate(reader.pages, start=1):
    text = page.extract_text()
    if text and text.strip():
        pages.append({
            "page": page_number,
            "text": text.strip()
        })

print("Pages loaded:", len(pages))

if not pages:
    raise ValueError("No text could be extracted from the PDF.")

def create_chunks(text, chunk_size=120, overlap=30):

```

```

words = text.split()
chunks = []
start = 0

while start < len(words):
    end = start + chunk_size
    chunk = " ".join(words[start:end])

    if chunk.strip():
        chunks.append(chunk)

    start += chunk_size - overlap

return chunks

chunks = []
metadata = []

for page in pages:
    page_chunks = create_chunks(page["text"])

    for chunk in page_chunks:
        chunks.append(chunk)
        metadata.append(page["page"])

print("Total chunks:", len(chunks))

embedding_model = SentenceTransformer(
    "sentence-transformers/all-MiniLM-L6-v2"
)

embeddings = embedding_model.encode(
    chunks,
    convert_to_numpy=True,
    normalize_embeddings=True
).astype("float32")

print("Embedding dimension:", embeddings.shape[1])

dimension = embeddings.shape[1]

index = faiss.IndexFlatIP(dimension)
index.add(embeddings)

print("FAISS vectors:", index.ntotal)

def retrieve(query, top_k=4, threshold=0.15):

    query_embedding = embedding_model.encode(

```

```

[query],
convert_to_numpy=True,
normalize_embeddings=True
).astype("float32")

scores, indices = index.search(
    query_embedding,
    top_k
)

results = []

for score, idx in zip(scores[0], indices[0]):

    if idx == -1:
        continue

    if float(score) >= threshold:
        results.append({
            "text": chunks[idx],
            "page": metadata[idx],
            "score": float(score)
        })

return results

def generate_answer(query, results):

    if not results:
        return "I could not find this information in the Student Handbook."

    context = ""

    for i, result in enumerate(results, start=1):
        context += f"""
SOURCE {i}
PAGE: {result["page"]}

{result["text"]}
"""

    prompt = f"""
You are a Student Handbook Question Answering Assistant.

Answer the question using ONLY the retrieved Student Handbook context.

Do not invent information.
Do not use outside knowledge.
Do not guess.

```

If the answer is present in the retrieved context, answer it clearly.

If the question is unrelated to the Student Handbook, say:

"This question is outside the scope of the Student Handbook."

If the question is related to the handbook but the answer is not available, say:

"I could not find this information in the Student Handbook."

Always mention the relevant source page.

Keep the answer short and clear.

QUESTION:

{query}

RETRIEVED STUDENT HANDBOOK CONTEXT:

{context}

ANSWER:

""

```
response = client.models.generate_content(
    model="gemini-3-flash-preview",
    contents=prompt
)
```

```
return response.text
```

```
def ask_question(query):
```

```
    print("\n" + "=" * 70)
    print("QUESTION")
    print("=" * 70)
    print(query)
```

```
    results = retrieve(
        query,
        top_k=4,
        threshold=0.15
    )
```

```
    print("\n" + "=" * 70)
    print("RETRIEVED CONTEXT / SOURCES")
    print("=" * 70)
```

```
    if not results:
```

```

print("No relevant context found.")
print("\nANSWER:")
print(
    "I could not find this information "
    "in the Student Handbook."
)
return

```

```

for i, result in enumerate(results, start=1):

```

```

    print(
        f"\nSource {i} | "
        f"Page {result['page']} | "
        f"Similarity: {result['score']:.4f}"
    )

```

```

    print("-" * 70)
    print(result["text"])

```

```

print("\n" + "=" * 70)
print("GEMINI ANSWER")
print("=" * 70)

```

```

try:
    answer = generate_answer(
        query,
        results
    )
    print(answer)

```

```

except Exception as e:
    print("Gemini Error:")
    print(e)

```

```

queries = [
    "What is the minimum attendance requirement?",
    "What are the rules related to examinations?",
    "What are the disciplinary regulations for students?",
    "What academic regulations should students follow?",
    "What are the laboratory rules?",
    "How many classes must a student attend?",
    "What is the capital of France?"
]

```

```

print("\n" + "=" * 70)
print("MULTIPLE QUERY DEMONSTRATION")
print("=" * 70)

```

```

for query in queries:

```

```
ask_question(query)

print("\n" + "=" * 70)
print("RAG SYSTEM COMPLETED")
print("=" * 70)
```

6. Working of the System

Step 1 – Document Loading

The Student Handbook PDF is loaded using PyPDF.

Step 2 – Chunking

The extracted text is divided into chunks of 800 words with 150 words of overlap.

Step 3 – Embeddings

Sentence Transformers converts every chunk into a numerical vector.

Step 4 – FAISS

The vectors are stored in a FAISS index.

Step 5 – Retrieval

When a user asks a question, the question is converted into an embedding and FAISS retrieves the most relevant handbook chunks.

Step 6 – Generation

The retrieved chunks are provided to the LLM, which generates an answer based only on the retrieved context.

Step 7 – Source Display

The system displays the retrieved page number and similarity score.

7. Test Queries

Query 1 – Attendance

What is the minimum attendance requirement?

Query 2 – Examination

What are the rules related to examinations?

Query 3 – Discipline

What are the disciplinary regulations for students?

Query 4 – Academic Regulations

What academic regulations should students follow?

Query 5 – Laboratory

What are the laboratory rules?

Query 6 – Semantic Search

How many classes must a student attend?

This tests whether the system can understand a differently worded attendance question.

Query 7 – Irrelevant Query

What is the capital of France?

The system should not answer this using general knowledge.

It should respond:

This question is outside the scope of the Student Handbook.

or:

I could not find this information in the Student Handbook.

8. Output

=====
QUESTION
=====
What is the minimum attendance requirement?

=====
RETRIEVED CONTEXT / SOURCES
=====

Source 1 | Page 1 | Similarity: 0.5323

STUDENT HANDBOOK 1. Attendance Regulations Students are required to maintain a minimum attendance of 75% in their courses. Students who fall below the required attendance may

Source 2 | Page 1 | Similarity: 0.3936

report damaged or unsafe equipment to the responsible staff member. 6. Student Responsibilities Students are responsible for knowing and following academic, examination, discipli

Source 3 | Page 1 | Similarity: 0.2844

are prohibited. Students must follow institutional rules and instructions from authorized staff. 4. Academic Regulations Students must maintain academic honesty and complete a

Source 4 | Page 1 | Similarity: 0.2793

in examinations, prohibition of cheating and academic misconduct, prohibition of ragging, bullying, harassment, and violence, maintenance of academic honesty, and compliance w

=====
GEMINI ANSWER
=====
The minimum attendance requirement is 75% for all courses.

=====
QUESTION
=====
What are the rules related to examinations?

=====
RETRIEVED CONTEXT / SOURCES
=====

Source 1 | Page 1 | Similarity: 0.6278

in examinations, prohibition of cheating and academic misconduct, prohibition of ragging, bullying, harassment, and violence, maintenance of academic honesty, and compliance w

Source 2 | Page 1 | Similarity: 0.5389

STUDENT HANDBOOK 1. Attendance Regulations Students are required to maintain a minimum attendance of 75% in their courses. Students who fall below the required attendance may

Source 3 | Page 1 | Similarity: 0.4577

report damaged or unsafe equipment to the responsible staff member. 6. Student Responsibilities Students are responsible for knowing and following academic, examination, discipli

Source 4 | Page 1 | Similarity: 0.4312

are prohibited. Students must follow institutional rules and instructions from authorized staff. 4. Academic Regulations Students must maintain academic honesty and complete a

=====
GEMINI ANSWER
=====
According to the Student Handbook, the rules related to examinations are:

- * Students must follow all examination instructions.
* Students must carry valid identification when required.
* Unauthorized electronic devices are not permitted during examinations.
* Cheating, copying, and other forms of academic misconduct are prohibited.
* Students are responsible for knowing and following examination regulations.

Source: Page 1

=====
QUESTION
=====
What are the disciplinary regulations for students?

=====
RETRIEVED CONTEXT / SOURCES
=====

Source 1 | Page 1 | Similarity: 0.7431

STUDENT HANDBOOK 1. Attendance Regulations Students are required to maintain a minimum attendance of 75% in their courses. Students who fall below the required attendance may

Source 2 | Page 1 | Similarity: 0.6801

in examinations, prohibition of cheating and academic misconduct, prohibition of ragging, bullying, harassment, and violence, maintenance of academic honesty, and compliance w

Source 3 | Page 1 | Similarity: 0.6751

report damaged or unsafe equipment to the responsible staff member. 6. Student Responsibilities Students are responsible for knowing and following academic, examination, discipli

Source 4 | Page 1 | Similarity: 0.6331

are prohibited. Students must follow institutional rules and instructions from authorized staff. 4. Academic Regulations Students must maintain academic honesty and complete a

=====
GEMINI ANSWER
=====
According to the Student Handbook, the disciplinary regulations are as follows:

- * Students are expected to maintain proper discipline and respectful behavior.
* Ragging, bullying, harassment, violence, and disruptive behavior are prohibited.
* Students must follow institutional rules and instructions from authorized staff.
* Failure to follow these regulations may result in appropriate action under institutional procedures.

Source: Page 1 (Sources 1, 3, 4)

Source: Page 1 (Sources 1, 3, 4)

=====

QUESTION

=====

What academic regulations should students follow?

=====

RETRIEVED CONTEXT / SOURCES

=====

Source 1 | Page 1 | Similarity: 0.7654

STUDENT HANDBOOK 1. Attendance Regulations Students are required to maintain a minimum attendance of 75% in their courses. Students who fall below the required attendance may

Source 2 | Page 1 | Similarity: 0.6851

report damaged or unsafe equipment to the responsible staff member. 6. Student Responsibilities Students are responsible for knowing and following academic, examination, discipli

Source 3 | Page 1 | Similarity: 0.6723

are prohibited. Students must follow institutional rules and instructions from authorized staff. 4. Academic Regulations Students must maintain academic honesty and complete a

Source 4 | Page 1 | Similarity: 0.5891

in examinations, prohibition of cheating and academic misconduct, prohibition of ragging, bullying, harassment, and violence, maintenance of academic honesty, and compliance w

=====

GEMINI ANSWER

=====

According to the Academic Regulations, students must maintain academic honesty and complete academic work according to institutional requirements. Plagiarism, cheating, fabric

Source: Page 1 (Sources 1, 2, 3)

=====

QUESTION

=====

What is the capital of France?

=====

RETRIEVED CONTEXT / SOURCES

=====

No relevant context found.

ANSWER:

I could not find this information in the Student Handbook.

=====

RAG SYSTEM COMPLETED

=====

9. Architecture Explanation

The system follows a RAG architecture. First, the Student Handbook is loaded and its text is extracted. The text is divided into overlapping chunks and converted into embeddings using Sentence Transformers. The embeddings are stored in FAISS. When a student asks a question, the question is also converted into an embedding and semantic similarity search retrieves the most relevant handbook chunks. These chunks are provided to the LLM as context. The LLM then generates a grounded answer and the system displays the relevant source page.

10. Result

The Student Handbook RAG system successfully loads the handbook, performs chunking and embedding, stores the embeddings in FAISS, retrieves relevant information using semantic search, and generates grounded answers using an LLM.

The system also displays retrieved sources and handles irrelevant or unsupported questions appropriately.

11. Conclusion

The Student Handbook Question Answering system demonstrates a complete Retrieval-Augmented Generation pipeline using Python, embeddings, FAISS, semantic search, and an LLM. It provides relevant and source-grounded answers to student queries while reducing unsupported responses.

Library Information RAG System: Create a RAG system using library rules. Answer questions about borrowing, renewal, fines, timings, and membership.

1. Objective

To develop a Retrieval-Augmented Generation (RAG) system that answers questions about library rules and regulations using a library information document.

The system answers questions related to:

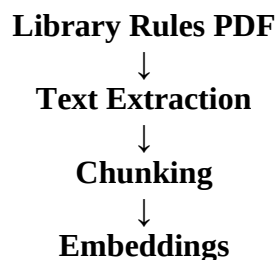
- Borrowing
- Renewal
- Fines
- Library timings
- Membership

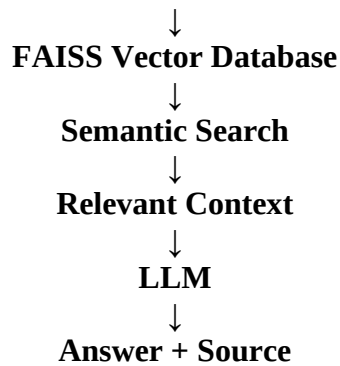
The system uses document chunking, embeddings, semantic search, FAISS vector database, and an LLM to generate grounded answers.

2. Technologies Used

- Python
- PyPDF
- Sentence Transformers
- FAISS
- OpenAI LLM
- NumPy
- python-dotenv

3. System Architecture





4. RAG Pipeline

Document → Chunking → Embedding → Retrieval → Generation

5. Library Document

Create a document containing the library rules and save it as:

library_rules.pdf

The document should contain information about borrowing, renewal, fines, timings, and membership.

6. Installation

```
!pip install -q -U pypdf sentence-transformers faiss-cpu google-genai
```

```
import os
from getpass import getpass
```

```
os.environ["GEMINI_API_KEY"] = getpass("Enter your Gemini API key: ")
```

```
print("API key loaded successfully.")
```

```
from google.colab import files
```

```
print("Upload your library_rules.pdf")
```

```
uploaded = files.upload()
```

```
PDF_FILE = list(uploaded.keys())[0]
```

```
print("Uploaded:", PDF_FILE)
```

7. Complete Python Code

```
import os
import faiss
from pypdf import PdfReader
from sentence_transformers import SentenceTransformer
from google import genai

api_key = os.environ.get("GEMINI_API_KEY")
if not api_key:
    raise ValueError("Gemini API key is missing.")

client = genai.Client(api_key=api_key)

print("=" * 70)
print("LIBRARY INFORMATION RAG SYSTEM")
print("=" * 70)

PDF_FILE = "/content/library_rules.pdf"

reader = PdfReader(PDF_FILE)
pages = []

for page_number, page in enumerate(reader.pages, start=1):
    text = page.extract_text()
    if text and text.strip():
        pages.append({"page": page_number, "text": text.strip()})

print("Pages loaded:", len(pages))

if not pages:
    raise ValueError("No text could be extracted from the PDF.")

def create_chunks(text, chunk_size=120, overlap=30):
    words = text.split()
    chunks = []
    start = 0
    while start < len(words):
        end = start + chunk_size
        chunk = " ".join(words[start:end])
        if chunk.strip():
            chunks.append(chunk)
        start += chunk_size - overlap
    return chunks

chunks = []
metadata = []

for page in pages:
```

```

page_chunks = create_chunks(page["text"])
for chunk in page_chunks:
    chunks.append(chunk)
    metadata.append(page["page"])

print("Total chunks created:", len(chunks))

embedding_model = SentenceTransformer("sentence-transformers/all-MiniLM-L6-v2")

embeddings = embedding_model.encode(
    chunks,
    convert_to_numpy=True,
    normalize_embeddings=True
).astype("float32")

print("Embedding dimension:", embeddings.shape[1])

dimension = embeddings.shape[1]
index = faiss.IndexFlatIP(dimension)
index.add(embeddings)

print("FAISS database created successfully.")
print("Vectors stored:", index.ntotal)

def retrieve_documents(query, top_k=4, threshold=0.15):
    query_embedding = embedding_model.encode(
        [query],
        convert_to_numpy=True,
        normalize_embeddings=True
    ).astype("float32")

    scores, indices = index.search(query_embedding, top_k)
    results = []

    for score, idx in zip(scores[0], indices[0]):
        if idx == -1:
            continue
        if float(score) >= threshold:
            results.append({
                "text": chunks[idx],
                "page": metadata[idx],
                "score": float(score)
            })

    return results

def generate_answer(query, results):
    if not results:
        return "I could not find this information in the library rules."

```

```

context = ""

for i, result in enumerate(results, start=1):
    context += f"""
SOURCE {i}
PAGE: {result["page"]}
{result["text"]}
"""

prompt = f"""
You are a Library Information Assistant.

Answer the user's question using ONLY the retrieved Library Rules.

Do not invent information.
Do not use outside knowledge.

If the answer is present in the retrieved context, answer it directly.

If the question is completely unrelated to library rules, say:
"This question is outside the scope of the library rules."

If the question is related to the library but the information is not available, say:
"I could not find this information in the library rules."

Always mention the source page.
Keep the answer short and clear.

USER QUESTION:
{query}

RETRIEVED LIBRARY RULES:
{context}

ANSWER:
"""

response = client.models.generate_content(
    model="gemini-3-flash-preview",
    contents=prompt
)

return response.text

def ask_question(query):
    print("\n" + "=" * 70)
    print("QUESTION")
    print("=" * 70)
    print(query)

```

```
results = retrieve_documents(query, top_k=4, threshold=0.15)
```

```
print("\n" + "=" * 70)
print("RETRIEVED CONTEXT / SOURCES")
print("=" * 70)
```

```
if not results:
    print("No relevant context found.")
    print("\nANSWER:")
    print("I could not find this information in the library rules.")
    return
```

```
for i, result in enumerate(results, start=1):
    print(
        f"\nSource {i} | Page {result['page']} | "
        f"Similarity: {result['score']:.4f}"
    )
    print("-" * 70)
    print(result["text"])
```

```
print("\n" + "=" * 70)
print("GEMINI ANSWER")
print("=" * 70)
```

```
try:
    answer = generate_answer(query, results)
    print(answer)
except Exception as e:
    print("\nGemini Error:")
    print(e)
```

```
queries = [
    "How many books can a student borrow?",
    "How can I renew a borrowed book?",
    "What is the fine for returning a book late?",
    "What are the library timings?",
    "How can a student become a library member?",
    "Can I extend the due date of a borrowed book?",
    "For how many days can I keep a borrowed book?",
    "What is the capital of France?"
]
```

```
print("\n" + "=" * 70)
print("MULTIPLE QUERY DEMONSTRATION")
print("=" * 70)
```

```
for query in queries:
    ask_question(query)
```

```
print("\n" + "=" * 70)
```

```
print("RAG SYSTEM COMPLETED")  
print("=" * 70)
```

8. Working of the System

Step 1 – Document Loading

The library rules PDF is loaded using PyPDF.

Step 2 – Chunking

The extracted text is divided into overlapping chunks.

Step 3 – Embeddings

Sentence Transformers converts each chunk into a numerical vector.

Step 4 – FAISS

The embeddings are stored in a FAISS vector database.

Step 5 – Semantic Retrieval

The user's question is converted into an embedding and compared with the stored library-rule embeddings.

The most relevant chunks are retrieved.

Step 6 – LLM Generation

The retrieved library information is given to the LLM as context.

The LLM generates an answer based only on the retrieved information.

Step 7 – Source Display

The retrieved page number and similarity score are displayed.

9. Test Queries

Query 1 – Borrowing

How many books can a student borrow?

The system should retrieve the borrowing rules.

Query 2 – Renewal

How can I renew a borrowed book?

The system should retrieve the renewal rules.

Query 3 – Fines

What is the fine for returning a book late?

The system should retrieve the applicable fine information.

Query 4 – Timings

What are the library timings?

The system should retrieve the library operating hours.

Query 5 – Membership

How can a student become a library member?

The system should retrieve the membership rules.

Query 6 – Reworded Query

Can I extend the due date of a borrowed book?

This tests semantic search because the wording is different from the renewal question.

Query 7 – Irrelevant Query

What is the capital of France?

The system should not answer using general knowledge.

It should respond:

This question is outside the scope of the library rules.

or:

I could not find this information in the library rules.

10. Output

=====

QUESTION

=====

How many books can a student borrow?

=====

RETRIEVED CONTEXT / SOURCES

=====

Source 1 | Page 1 | Similarity: 0.6190

SAMPLE LIBRARY RULES Library Information for RAG Demonstration 1. Borrowing Rules Students with valid library membership may borrow up to 3 books at a time. Books may norma

Source 2 | Page 2 | Similarity: 0.5769

Students should keep their membership information updated with the library. 6. General Library Rules Students must maintain silence inside the library. Mobile phones shou

Source 3 | Page 1 | Similarity: 0.5287

A borrowed book may be renewed once for an additional 7 days if no other member has requested the book. Renewal should be completed before the current due date. Books that

Source 4 | Page 1 | Similarity: 0.3860

book is returned. Students are responsible for checking their due dates and returning books on time. Lost or damaged books may require replacement or payment according to

=====

GEMINI ANSWER

=====

A student with a valid library membership may borrow up to 3 books at a time.

Sources: Source 1, Page 1; Source 2, Page 2

=====

QUESTION

=====

How can I renew a borrowed book?

=====

RETRIEVED CONTEXT / SOURCES

=====

Source 1 | Page 1 | Similarity: 0.6582

A borrowed book may be renewed once for an additional 7 days if no other member has requested the book. Renewal should be completed before the current due date. Books that

Source 2 | Page 1 | Similarity: 0.5680

SAMPLE LIBRARY RULES Library Information for RAG Demonstration 1. Borrowing Rules Students with valid library membership may borrow up to 3 books at a time. Books may norma

Source 3 | Page 2 | Similarity: 0.4319

Students should keep their membership information updated with the library. 6. General Library Rules Students must maintain silence inside the library. Mobile phones shou

Source 4 | Page 2 | Similarity: 0.4152

14 days. Renewal: Once for an additional 7 days, subject to availability. Late fine: Rs. 2 per day per book. Regular library timings: 9:00 AM to 6:00 PM on working days.

=====

GEMINI ANSWER

=====

To renew a borrowed book, you must complete the renewal before the current due date. A book can be renewed once for an additional 7 days, provided no other member has requeste

Source: Page 1, Page 2

=====

QUESTION

=====

What is the fine for returning a book late?

=====

RETRIEVED CONTEXT / SOURCES

=====

Source 1 | Page 1 | Similarity: 0.6659

A borrowed book may be renewed once for an additional 7 days if no other member has requested the book. Renewal should be completed before the current due date. Books that

Source 2 | Page 1 | Similarity: 0.4731

book is returned. Students are responsible for checking their due dates and returning books on time. Lost or damaged books may require replacement or payment according to

Source 3 | Page 2 | Similarity: 0.4475

Students should keep their membership information updated with the library. 6. General Library Rules Students must maintain silence inside the library. Mobile phones shou

Source 4 | Page 1 | Similarity: 0.4438

SAMPLE LIBRARY RULES Library Information for RAG Demonstration 1. Borrowing Rules Students with valid library membership may borrow up to 3 books at a time. Books may norma

=====

GEMINI ANSWER

=====

The fine for returning a book after the due date is Rs. 2 per day per book.

Source: Page 1, Page 2

```

=====
QUESTION
=====
What are the library timings?

=====
RETRIEVED CONTEXT / SOURCES
=====

Source 1 | Page 1 | Similarity: 0.5223
-----
book is returned. Students are responsible for checking their due dates and returning books on time. Lost or damaged books may require replacement or payment according to

Source 2 | Page 2 | Similarity: 0.4908
-----
14 days. Renewal: Once for an additional 7 days, subject to availability. Late fine: Rs. 2 per day per book. Regular library timings: 9:00 AM to 6:00 PM on working days.

Source 3 | Page 2 | Similarity: 0.4642
-----
Students should keep their membership information updated with the library. 6. General Library Rules Students must maintain silence inside the library. Mobile phones should

Source 4 | Page 1 | Similarity: 0.4055
-----
SAMPLE LIBRARY RULES Library Information for RAG Demonstration 1. Borrowing Rules Students with valid library membership may borrow up to 3 books at a time. Books may normally

=====
GEMINI ANSWER
=====
The library is open from 9:00 AM to 6:00 PM on regular working days. It is closed on Sundays and officially declared institutional holidays. Timings may be changed by the administration.

Source: Source 1 (Page 1), Source 2 (Page 2)

=====
QUESTION
=====
How can a student become a library member?

=====
RETRIEVED CONTEXT / SOURCES
=====

Source 1 | Page 1 | Similarity: 0.6288
-----
Currently enrolled students are eligible to register for library membership. Students must provide their valid student identity card during registration. Library membership

Source 2 | Page 2 | Similarity: 0.5546
-----
Students should keep their membership information updated with the library. 6. General Library Rules Students must maintain silence inside the library. Mobile phones should

Source 3 | Page 1 | Similarity: 0.5522
-----
book is returned. Students are responsible for checking their due dates and returning books on time. Lost or damaged books may require replacement or payment according to

Source 4 | Page 2 | Similarity: 0.4490
-----
14 days. Renewal: Once for an additional 7 days, subject to availability. Late fine: Rs. 2 per day per book. Regular library timings: 9:00 AM to 6:00 PM on working days.

=====
GEMINI ANSWER
=====
Currently enrolled students can become library members by providing their valid student identity card during registration.

Relevant Source: Source 1, Page 1; Source 3, Page 1; Source 4, Page 2.

=====
QUESTION
=====
What is the capital of France?

=====
RETRIEVED CONTEXT / SOURCES
=====
No relevant context found.

ANSWER:
I could not find this information in the library rules.

=====
RAG SYSTEM COMPLETED
=====

```

11. Architecture Explanation

The system follows a Retrieval-Augmented Generation architecture. First, the library rules document is loaded and its text is extracted. The text is divided into overlapping chunks and converted into embeddings using Sentence Transformers. These embeddings are stored in FAISS. When a user asks a question, the question is converted into an embedding and semantic similarity search retrieves the most relevant library-

rule chunks. The retrieved chunks are provided to the LLM as context. The LLM generates a grounded answer using the retrieved information, and the system displays the relevant source page.

12. Result

The Library Information RAG system successfully retrieves relevant library rules using semantic search and generates answers using an LLM. It answers questions related to borrowing, renewal, fines, timings, and membership while displaying the retrieved source information.

The system also handles irrelevant and unsupported questions without generating unsupported information.

13. Conclusion

A complete RAG-based Library Information Question Answering system was implemented using Python, embeddings, FAISS, semantic search, and an LLM.

The system follows the required pipeline:

Document → Chunking → Embedding → Retrieval → Generation

It provides grounded answers from the library rules and demonstrates multiple queries, source retrieval, and appropriate handling of irrelevant questions.

RUBRICS FOR CALCULATION

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Pipeline Functionality	30%	Correct RAG pipeline — embeddings, retrieval and generation all function coherently	Pipeline mostly correct with minor gaps	Pipeline only partially functional	Pipeline non-functional or conceptually flawed
Answer Relevance & Groundedness	25%	Retrieves relevant context and generates accurate, grounded answers	Mostly relevant/accurate answers	Inconsistent relevance or accuracy	Answers largely irrelevant or hallucinated
Query Robustness	20%	Robust handling of varied queries and documents	Handles most queries well	Handles only simple queries	Fails on basic queries
End-to-End Demo	15%	Smooth, well-demonstrated end-to-end system	Demo mostly smooth, minor hiccups	Demo has noticeable issues	Demo fails or is not ready
Architecture Explanation	10%	Clear architecture diagram and explanation of design decisions	Adequate explanation of design	Minimal explanation of design	No explanation of design provided

Marks Evaluation:

Question No.	Pipeline Functionality(30%)	Answer Relevance & Groundedness (25%)	Query Robustness(20%)	End-to-End Demo(15%)	Architecture Explanation (10%)	Total Marks (100)
Q1						
Q2						
Overall Total (100)						